

EDA INSIGHT MEMO

Project FORESIGHT — Demand & Inventory Intelligence

Client: NorthBay Living | Zidio Development Internship | August 2026

Dataset: retail_forecasting_cleaned_v2 | 365,000 daily records | 200 SKUs

1. Purpose of this Memo

This memo summarises the exploratory analysis of NorthBay Living's sales and inventory history. It highlights demand patterns, stockout and overstock signals, and the business implications that should guide forecasting and inventory decisions.

2. Data Snapshot

Metric	Value
Date range	2021-08-14 → 2026-08-12 (~5 years)
Records / SKUs	365,000 daily rows / 200 SKUs
Categories	Decor, Small Appliances, Furniture
Total units sold	~2.01 million
Total revenue	~₹616.7 crore (approx.)

3. Key Demand Patterns

3.1 Trend & Seasonality

Monthly revenue shows a clear overall upward trajectory across the five-year window, with visible seasonal peaks. Average revenue by calendar month confirms seasonality; demand is not flat. Weekly aggregation and lag-52 features will be important for forecasting.

3.2 Category & Subcategory Performance

Furniture and Small Appliances contribute the largest share of revenue. Within subcategories, sofas, certain appliances, and selected décor items (rugs, wall art) are the strongest contributors. Inventory policy should not treat all categories equally.

3.3 Revenue Concentration (80/20)

A minority of SKUs generate the large majority of revenue. This concentration means forecasting accuracy and stock availability on the top movers have outsized impact on sales and customer experience.

3.4 Promotion & Holiday Effects

Promotion days show higher average revenue than non-promotion days. Named events (Memorial Day Sale, Black Friday / Cyber Monday, Holiday Countdown, etc.) are the strongest lifts. Holidays also elevate average revenue. Promo and holiday flags are useful predictive features.

4. Inventory Risk Findings

4.1 Stockouts — The Dominant Problem

198 out of 200 SKUs hit zero on-hand inventory at least once. The most severe cases had 400–577 stockout days (e.g. SKU0127, SKU0192, SKU0049). On a non-trivial share of those days, the previous day had positive sales — a proxy for potential lost demand. Stockouts are frequent and material.

4.2 Dead Stock / Slow Movers (Last 90 Days)

Using a 90-day lookback, **zero SKUs** qualified as pure dead stock (0 recent sales + positive on-hand) and zero qualified as slow movers under the chosen thresholds. In the most recent quarter every SKU still recorded some sales. The immediate problem is therefore stockouts and availability, not capital locked in completely inactive items.

4.3 Inventory–Sales Relationship

On-hand inventory and sales are positively but moderately correlated. Reorder point correlates more strongly with both sales and on-order quantities, suggesting the current reorder rules partially reflect demand intensity but are not preventing frequent stockouts.

5. Business Insights for the Operations Team

Insight 1 — Stockouts are the primary risk, not dead stock.

198 of 200 SKUs have stocked out. Top offenders lost hundreds of selling days. Improving availability on these SKUs will protect revenue more than clearing non-existent dead stock.

Insight 2 — Focus forecasting and inventory effort on the revenue core.

A small set of SKUs drives most revenue. Prioritise weekly demand forecasts, safety stock, and reorder discipline on this core set before spreading effort evenly across all 200 SKUs.

Insight 3 — Promotions and holidays are reliable demand drivers.

Promo and holiday periods lift average revenue. The forecasting model and the planning calendar should explicitly account for these events so inventory is positioned ahead of known peaks.

Insight 4 — Furniture and Small Appliances need tighter control.

These categories generate substantial revenue yet appear frequently among high-stockout SKUs. Longer lead times in some of these items make early warning especially valuable.

Insight 5 — Current data supports a weekly SKU-level forecast + risk layer.

Five years of daily history, clear seasonality, promo signals, and inventory positions are sufficient to build a seasonal-naïve baseline and a stronger model, then convert forecasts into stockout/overstock risk scores.

6. Implications for the Next Stages

- Aggregate to weekly grain and engineer lag / rolling / calendar / promo features.
- Build a seasonal-naïve baseline and report WAPE; then beat it with a stronger model.
- Convert forecasts + inventory into stockout and overstock risk scores with recommended actions.
- Quantify rupee impact (sales at risk from stockouts, capital in any future overstock).

7. Conclusion

The exploratory analysis confirms that NorthBay Living's main inventory pain is repeated stockouts on a large share of the assortment, concentrated revenue in a minority of SKUs, and meaningful seasonality and promotion effects. There is little evidence of widespread dead stock in the latest 90 days. These findings directly support the engagement objectives: a weekly SKU-level demand forecast, an early-warning risk layer, and a usable planning view for the operations team.